Chromosomal Theory of Inheritance & Linkage

Worksheet · PU-II Biology, Chapter 4

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Worksheet – Chromosomal Theory of Inheritance & Linkage

15 questions · answer key on the last page

Q1. Who is known as the "father of experimental genetics"?

(a) Gregor Mendel
(b) Thomas Hunt Morgan
(c) Walter Sutton
(d) Alfred Sturtevant

Q2. Morgan's genetic experiments were primarily carried out on which organism?

(a) Garden pea
(b) Guinea pig
(c) Fruit fly (Drosophila melanogaster)
(d) Honey bee

Q3. Which two scientists proposed that the behaviour of chromosomes parallels the behaviour of genes?

(a) Mendel and Punnett
(b) Walter Sutton and Theodore Boveri
(c) Morgan and Sturtevant
(d) Landsteiner and Bernstein

Q4. Linkage refers to:

(a) The blending of two alleles in F1
(b) The physical association of genes on the same chromosome
(c) The independent assortment of genes on different chromosomes
(d) The mutation of a single gene

Q5. Recombination refers to:

(a) The physical association of genes on a chromosome
(b) The generation of non-parental (new) gene combinations
(c) The complete linkage of two genes
(d) The doubling of chromosome number

Q6. One map unit (m.u.), used in genetic mapping, is equivalent to:

(a) 1% recombination frequency between two linked genes
(b) 10% recombination frequency
(c) 100% recombination frequency
(d) The physical length of a chromosome

Q7. Who used gene recombination frequency to map the relative positions of genes on a chromosome?

(a) Thomas Hunt Morgan
(b) Alfred Sturtevant
(c) Theodore Boveri
(d) Reginald Punnett

Q8. Chromosomes other than the sex chromosomes are called:

(a) Allosomes
(b) Autosomes
(c) Homologues
(d) Alleles

Q9. Which of the following was NOT a reason Morgan chose Drosophila melanogaster for his genetic studies?

(a) It can be grown on synthetic medium in the laboratory
(b) Its life cycle completes in about two weeks

(c) It has a very long generation time of several years

(d) Males and females are easily distinguished

Q10. In Morgan's dihybrid cross experiment, the parental generation was:

(a) Yellow-bodied white-eyed females x brown-bodied red-eyed males

(b) Brown-bodied white-eyed females x yellow-bodied red-eyed males

(c) Yellow-bodied red-eyed females x brown-bodied white-eyed males

(d) Grey-bodied females x black-bodied males

Q11. In Morgan's Drosophila dihybrid cross, the F2 ratio obtained:

(a) Exactly matched the standard 9:3:3:1 Mendelian ratio

(b) Was significantly different from the standard 9:3:3:1 ratio

(c) Showed complete linkage with no recombinants

(d) Could not be determined due to sterility

Q12. Morgan explained the deviation from the expected dihybrid ratio in Drosophila by proposing that:

(a) The two genes assorted completely independently

(b) The two genes were located on the same (X) chromosome

(c) One gene was on an autosome and one on the Y chromosome

(d) The trait showed polygenic inheritance

Q13. During which phase of meiosis do homologous chromosome pairs separate randomly, forming the physical basis of independent assortment?

(a) Prophase I

(b) Metaphase I

(c) Anaphase I

(d) Telophase II

Q14. In Morgan's Cross A (genes for body colour "y" and eye colour "w"), the recombination frequency observed was:

(a) 1.3%

(b) 37.2%

(c) 50%

(d) 98.7%

Q15. In Morgan's Cross B (genes for eye colour "w" and wing size/miniature "m"), the recombination frequency observed was:

(a) 1.3%

(b) 37.2%

(c) 62.8%

(d) 98.7%

Answer Key & Explanations

Q1. (b) Thomas Hunt Morgan — Thomas Hunt Morgan, who worked extensively on Drosophila melanogaster, is called the father of experimental genetics and father of sex linkage.

Q2. (c) Fruit fly (Drosophila melanogaster) — Morgan used Drosophila melanogaster because it has a short life cycle, produces many progeny, and shows clear hereditary variations.

Q3. (b) Walter Sutton and Theodore Boveri — Walter Sutton and Theodore Boveri noted that chromosomes, like genes, occur in pairs and separate during gamete formation, forming the basis of the chromosomal theory.

Q4. (b) The physical association of genes on the same chromosome — Linkage is the physical association of genes located on the same chromosome, which causes them to be inherited together more often than expected.

Q5. (b) The generation of non-parental (new) gene combinations — Recombination is the generation of new, non-parental gene combinations, typically through crossing over during meiosis.

Q6. (a) 1% recombination frequency between two linked genes — One map unit (also called one centiMorgan) equals 1% crossing over/recombination frequency between two linked genes.

Q7. (b) Alfred Sturtevant — Alfred Sturtevant used recombination frequency as a measure of the physical distance between genes, creating the first genetic maps.

Q8. (b) Autosomes — Autosomes are all chromosomes other than the sex chromosomes (allosomes), which are involved in sex determination.

Q9. (c) It has a very long generation time of several years — Drosophila was chosen precisely because its life cycle is short (about two weeks), not because it has a long generation time, which would be a disadvantage.

Q10. (a) Yellow-bodied white-eyed females x brown-bodied red-eyed males — Morgan crossed yellow-bodied, white-eyed female Drosophila with brown-bodied, red-eyed males to study body colour and eye colour inheritance.

Q11. (b) Was significantly different from the standard 9:3:3:1 ratio — Morgan found the F2 ratio deviated significantly from the expected Mendelian dihybrid ratio, because the two genes involved were linked on the same chromosome.

Q12. (b) The two genes were located on the same (X) chromosome — Morgan explained the deviation by showing that the body colour and eye colour genes were both located on the X chromosome, so they did not assort independently.

Q13. (c) Anaphase I — During Anaphase I of meiosis, homologous chromosome pairs aligned at the metaphase plate separate randomly, providing the physical basis for independent assortment.

Q14. (a) 1.3% — Morgan's Cross A between the yellow (y) and white (w) genes showed 98.7% parental type and only 1.3% recombinant type, indicating tight linkage.

Q15. (b) 37.2% — Morgan's Cross B between white (w) and miniature (m) genes showed 62.8% parental type and 37.2% recombinant type, indicating looser linkage.